

Juniper Networks[®] vMX Release Notes

Release 15.1F4
22 July 2020
Revision 1

These release notes accompany this release of the Juniper Networks[®] virtual MX Series router (vMX). They describe new and changed features, limitations, and known problems in the software.

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Introduction

The virtual MX Series router (vMX) is an MX Series router optimized to run on x86 servers. We recommend Ubuntu as the host OS.

vMX allows you to leverage Junos OS Release 15.1 to provide quick and flexible deployment. vMX provides the following benefits:

- Optimizes carrier-grade routing for the x86 environment
- Simplifies operations by consistency with MX Series routers
- Introduces new services without reconfiguration of current infrastructure

This release of vMX supports most of the features available on Juniper Networks MX Series routers with the following exception:

- High availability features such as virtual Routing Engine redundancy are not supported in this release.

New and Changed Features

This section describes the new features and enhancements in this release.

- **ESXi hypervisor**—VMware ESXi 5.5 is supported as a hypervisor. You can use vSphere 5.5 or vCenter Server for the management client. E1000 and VMXNET3 are the supported physical PCI devices.
- **Flow caching**—For SR-IOV deployments, you can enable flow caching to improve performance. When you enable flow caching, make sure you have configured the proper number of vCPUs and memory for your VMs. You must also enable hyperthreading in BIOS.
- **Licenses**—Bandwidth licenses are additive. If you add multiple bandwidth licenses, you are entitled to use the sum of all the bandwidth licenses for capacity. In previous releases, you are entitled to use only the capacity of the largest bandwidth license when you add multiple bandwidth licenses.

Known Behavior

This section contains the known behaviors and limitations in this release.

- Scale limitation is observed with VLAN tag operation and circuit cross-connect (CCC).

- When vMX is deployed, the management port is not configured, so you must use the serial console for configuration. Only a small number of configuration lines can be pasted in the vMX console. As a workaround, perform initial configuration to set the root password and to allow SSH access in the vMX console and perform further configuration using SSH.

Known Issues

This section lists the known issues in this release.

- LACP packets are getting dropped on the bridge. [PR1059231](#)
- LLDP packets are getting dropped on the bridge (for virtio). [PR1066850](#)
- When the FPC is restarting, **COS(cos_chassis_scheduler_pre_add_action:2137)** error messages appear indicating that flexible queuing mode is not enabled. [PR1070655](#)
- Output for the **show pfe statistics traffic** command might take a few seconds to appear and many **MIB2D_COUNTER_DECREASING: pfes_stats_delta: counter** messages appear in the log file before the response time improves. [PR1071659](#)
- When the FPC is restarting, a **Received unsupported pic_mask 0x1 ignored message** message appears in the syslog file. [PR1072436](#)
- Traffic loss occurs at a remote receiver because of lost remote PIM joins to the local receiver. [PR1087031](#)
- The VMXNET3 driver does not support large MTU. [PR1120215](#)
- When committing the configuration, **%DAEMON-4-SNMP_EVLIB_FAILURE: PFED ran out of transfer credits with PFE.Failed to get stats.** messages sometimes appear in the syslog file. [PR1124902](#)
- When committing the configuration, **pfed: %USER-3: downward spike received from pfe for opackets_reply:3545 opackets_record:48932521** messages sometimes appear in the syslog file. [PR1146002](#)

System Requirements

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These topics provide system requirements for each supported environment.

Minimum Hardware and Software Requirements for KVM

[Table 1 on page 4](#) lists the hardware requirements.

Table 1: Minimum Hardware Requirements for KVM

Description	Value
Sample system configuration	For lab simulation and low performance (less than 100 Mbps) use cases, any x86 processor (Intel or AMD) with VT-d capability can be used. For all other use cases, Intel Ivy Bridge processors or later are required. Example of Ivy Bridge processor: Intel Xeon E5-2667 v2 @ 3.30 GHz 25 MB Cache For single root I/O virtualization (SR-IOV) NIC type, use Intel 82599-based PCI-Express cards (10 Gbps) and Ivy Bridge processors.
Number of cores	For lab simulation use case or low-bandwidth applications: 4 (1 for VCP and 3 for VFP) For high-bandwidth applications: 8 (1 for VCP and 7 for VFP)
Memory	For lab simulation use case or low-bandwidth applications: Minimum of 10 GB (2 GB for VCP, 8 GB for VFP) For high-bandwidth applications: Minimum of 16 GB (4 GB for VCP, 12 GB for VFP) Additional 2 GB recommended for host OS
Storage	Local or NAS
Other requirements	Intel VT-d capability Hyperthreading

[Table 2 on page 5](#) lists the software requirements.

Table 2: Minimum Software Requirements for KVM

Description	Value
Operating system	Ubuntu 14.04 LTS (recommended host OS) Linux 3.13.0-32-generic
Virtualization	QEMU-KVM 2.0.0+dfsg-2ubuntu1.11 or later
Required packages NOTE: Other additional packages might be required to satisfy all dependencies.	bridge-utils qemu-kvm libvirt-bin python python-netifaces vnc4server libyaml-dev python-yaml numactl libparted0-dev libpciaccess-dev libnuma-dev libyajl-dev libxml2-dev libglib2.0-dev libnl-dev libnl-dev python-pip python-dev libxml2-dev libxslt-dev NOTE: libvirt 1.2.19

Minimum Hardware and Software Requirements for VMware

Table 3 on page 5 lists the hardware requirements.

Table 3: Minimum Hardware Requirements for VMware

Description	Value
Number of cores	For performance mode: Minimum of 8 • 1 for VCP • 7 for VFP For lite mode (default): Minimum of 4 • 1 for VCP • 3 for VFP
Memory	For performance mode: Minimum of 16 GB • 4 GB for VCP • 12 GB for VFP For lite mode: Minimum of 10 GB • 2 GB for VCP • 8 GB for VFP
Storage	Local or NAS

Table 4 on page 6 lists the software requirements.

Table 4: Software Requirements for VMware

Description	Value
Hypervisor	ESXi 5.5 Update 2
Management Client	vSphere 5.5 or vCenter Server

Requesting Technical Support

Technical product support is available through the Juniper Networks Technical Assistance Center (JTAC). If you are a customer with an active J-Care or JNASC support contract, or are covered under warranty, and need post-sales technical support, you can access our tools and resources online or open a case with JTAC.

- JTAC policies—For a complete understanding of our JTAC procedures and policies, review the *JTAC User Guide* located at <https://www.juniper.net/us/en/local/pdf/resource-guides/7100059-en.pdf> .
- Product warranties—For product warranty information, visit <https://www.juniper.net/support/warranty/> .
- JTAC hours of operation—The JTAC centers have resources available 24 hours a day, 7 days a week, 365 days a year.

Self-Help Online Tools and Resources

For quick and easy problem resolution, Juniper Networks has designed an online self-service portal called the Customer Support Center (CSC) that provides you with the following features:

- Find CSC offerings: <https://www.juniper.net/customers/support/>
- Search for known bugs: <http://www2.juniper.net/kb/>
- Find product documentation: <https://www.juniper.net/documentation/>
- Find solutions and answer questions using our Knowledge Base: <https://kb.juniper.net/>
- Download the latest versions of software and review release notes: <https://www.juniper.net/customers/csc/software/>

- Search technical bulletins for relevant hardware and software notifications:
<https://www.juniper.net/alerts/>
- Join and participate in the Juniper Networks Community Forum:
<https://www.juniper.net/company/communities/>
- Open a case online in the CSC Case Management tool: <https://www.juniper.net/cm/>

To verify service entitlement by product serial number, use our Serial Number Entitlement (SNE) Tool:
<https://tools.juniper.net/SerialNumberEntitlementSearch/>

Opening a Case with JTAC

You can open a case with JTAC on the Web or by telephone.

- Use the Case Management tool in the CSC at <https://www.juniper.net/cm/> .
- Call 1-888-314-JTAC (1-888-314-5822 toll-free in the USA, Canada, and Mexico).

For international or direct-dial options in countries without toll-free numbers, see
<https://www.juniper.net/support/requesting-support.html> .

Revision History

22 July 2020—Revision 2, Junos OS Release 15.1F4—vMX

30 December 2015—Revision 1, Junos OS Release 15.1F4—vMX

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